Unexpectedly low marine heatwave forecast skill during the 2023–2024 El Niño

Supplementary Methods and data coverage

Scope of the supplementary analyses

These analyses address three questions: whether MHW forecast performance differs from historical expectations; how tropical and remote atmospheric forecasts differ; and how regional SST associations enter the diagnostic allocation of the regional map difference. Figures 1–4 and 16–17 assess MHW intensity, occurrence and sampling coverage; Figs. 5–11 examine tropical and remote atmospheric forecasts; Figs. 12–15 evaluate regional SST associations, their temporal transfer, a single-model process comparison and the signed map accounting. Supplementary Table 1 identifies the data and sampling support of each analysis.

Events, repeated samples and uncertainty

The earlier strong events are 1997–1998 and 2015–2016; the focal event is 2023–2024. The nine-event amplitude regression additionally uses 1991–1992, 1994–1995, 2002–2003, 2004–2005, 2006–2007, 2009–2010 and 2018–2019. Month, lead, model and spatial-grid replication do not create additional independent ENSO events. In Fig. 2, the same 2,000 resampled calendar-year blocks are used jointly for both phases, yielding percentile intervals for correlations and their slope contrast. Prediction intervals in Fig. 1 describe a new event under the linear regression assumptions. Lead and historical-reference ranges elsewhere are descriptive ranges, not confidence intervals. No new multiple-comparison significance claims are made.

AUC and conditional intensity errors

Global AUC in main Fig. 1a,b remains the original monthly spatial discrimination metric. The spatial sensitivity analyses in Supplementary Figs. 1c and 16–17 pool raw cell-month samples within an 11 × 11 array of 1° ocean grid cells (centre ±5 grid centres), restricted to the same major ocean basin and 60°S–60°N. Longitude wraps periodically. The primary window is April of the event year through September of the next year (18 months). Forecast records from the original July–June window are reused without modification; additional months use the same archived reader. Neighbour cells must have complete paired records for all events and leads. For each event and lead, weighted ROC AUC ranks forecast SST minus its local calendar-month Q90 against observed SST above Q90, with cosine-latitude sample weights and half credit for ties. A pool with only one observed class is undefined. Canonical-minus-target differences use equal weights for the two historical events and nine leads; only centres with all 27 finite AUCs enter the map. No interpolation or smoothing of the computed AUC is applied. Overlapping neighbourhoods reduce effective spatial independence despite the unchanged 1° output grid. Figure 17 separately checks 12 versus 18 months and single-cell versus 11° support. Forecast/observed intensity is max(SST − Q90,0); squared errors are pooled on observed-MHW samples only, using the same neighbourhood and time support. Roots follow MSE aggregation, including equal-event canonical averaging and region averaging. Regional neighbourhood metrics cannot be summed to recover the original global AUC gap. No claim of increased independent sample size is made.

Forecast extraction shared with the main SST figures

Figures 1–4 and 15–17 retain the archived SST extraction used in main Figs. 1 and 4. That extraction averages stored initialization coordinates in some selected forecast files. A file-level audit has identified cases where filename-based and exact initialization-coordinate selections differ, including GFDL files for September and December 2023 with conflicting source-coordinate attribution. Consequently these supplementary SST panels describe the retained analysis, not an independently initialization-controlled validation.
Exact-coordinate extraction changes the regression-based global deficit in the separately available sensitivity analysis; it is not silently substituted here. The magnitude of the SST-derived skill differences is therefore conditional on this extraction rule.

Tropical correlation comparisons

a,c,e, Cosine-latitude-weighted spatial correlations between forecast and observed DJF anomalies for SST, zonal wind stress and convection, represented by precipitation. All variables use NASA GEOS-S2S, the same four ensemble members (1–4), the same target months and leads, and the same tropical Pacific ocean mask (20°S–20°N, 140°E–100°W; 4,833 grid cells). The two historical strong El Niño events are shown separately. b,d,f, Relative correlation decrease: 100 × (r_reference − r_2023–2024)/r_reference, using each historical event and their equal-event mean. Positive values indicate deterioration; negative values indicate improvement. All right-hand panels share the same vertical scale. Lead 9 is unavailable and remains blank/NA; no nearest-lead substitution is used. Forecast initialization equals the verifying month minus the labelled lead; the internal S coordinate must match that initialization and L must equal lead + 0.5. Monthly anomalies are calculated against 1992–2020, with separate lead-dependent model and observed climatologies using identical available calendar months. DJF is averaged before spatial correlation. The same ocean mask is used for all 72 valid event–variable–lead correlations. Eight leads within each of three events are repeated evaluations, not independent events; reference curves are sensitivity comparisons, not confidence intervals. This is a controlled single-model comparison, not a multimodel attribution of MHW forecast-skill loss. Across eight leads, mean correlations change from 0.933 to 0.668 for SST, from 0.764 to 0.273 for zonal wind stress and from 0.810 to 0.612 for convection. Thus this matched comparison supports particularly strong wind-stress deterioration, but does not show greater deterioration in both atmospheric variables than in SST. The earlier mixed-protocol figure is retained in the preceding release, not used for this strict comparison.

Exact-coordinate DJF circulation comparisons

Figures 6–11 use DJF targets and exact available future leads 1–8 for COLA–RSMAS–CCSM4 (members 1–10), NASA–GEOS-S2S (1–4) and GFDL–SPEAR (1–15). Equal weights are applied to members within a model, then to model fields. All required fields must be finite. NCEP–CFSv2 is not in this fixed source set because its 1997–1998 Z200 product is unavailable there; NASA limits the common exact lead range to 1–8. Each calendar-month climatology spans 1992–2020, excluding February 2018 uniformly after the corresponding NASA field failed the availability check, leaving 29 December, 29 January and 28 February samples. No target-event DJF month is removed. The spatial correlation centers each field before cosine-latitude-weighted covariance; RMSE retains amplitude and sign-independent error magnitude. Pooled tropical–remote associations include between-event differences; their within-event counterparts cannot be substituted for them.

Region definitions

For main Figs. 3–4 and Supplementary Figs. 7 and 12–15, NOAA WOA major-basin identity is intersected with latitude bands: TP/TA, 23.5°S–23.5°N; NP/NA, 23.5°N–60°N; SP/SA, 60°S–23.5°S. IO spans the Indian Ocean from 60°S to its northern boundary. Grid-centre membership assigns −23.5° to the tropical band and +23.5° to the northern band, with no overlap. NP, SP, TA, NA, SA and IO are the six atmospheric recipient ocean masks. Global circulation comparisons additionally include land. The spatial sensitivity analyses in Supplementary Figs. 1c and 16–17 use the retained ±20° split; they are shown explicitly as a different support, not silently relabelled as the process regions.

Historical SST association and reconstruction

Figures 12–13 use regional ERSST SST and ORAS5 mixed-layer temperature, surface heating and horizontal advection. The fitting record is January 1985–December 2020 (432 months). ENSO is a separate group; IOD, PDO, NAO and SAM form the other-mode group. Local variables are previous-month SST and mixed-layer temperature, contemporaneous net surface heating divided by mixed-layer heat capacity, and horizontal advection. Full-sample month effects and linear trend are removed for the in-sample variance allocation. Group Shapley values average marginal explained variance over group orders; nested Owen values split the local parent while preserving group membership. The unresolved fraction is 1 − full-model R^2. All shares sum to one. For forward reconstruction, the seasonal cycle, trend, predictor scaling and regression are trained only before each test block. Test-time contemporaneous observed predictors make this diagnostic reconstruction rather than an actionable forecast.

CFSv2-only process check

Figure 14 uses a separately retrieved CFSv2 process archive, four members (00, 06, 12 and 18), DJF and leads 1, 3 and 6 months. Verification uses ERA5 upper-level circulation and ORAS5 net ocean surface heat flux. The circulation quantity is the departure from contemporaneous zonal mean, not the calendar-month anomaly used in main Fig. 3. Net surface flux is positive downward into the ocean. Squared errors are averaged across the three months and cosine-latitude-weighted common ocean cells before taking the root. Each RMSE is divided by the mean of the two earlier-event RMSEs at the same region and lead. The regional masks have been recomputed at ±23.5°. CFSv2 is also one component of the main SST ensemble; this is therefore a separately processed single-model consistency check, not a model-independent validation of the mechanism.

Exact relationship between Supplementary Fig. 15 and main Fig. 4

Let $d(g) = 0.7050185255 - \mathrm{AUC}(g)$, where AUC(g) is the nine-lead mean of the local 18-month, 11°-neighbourhood AUC, and let a(g) be cosine-latitude area weights normalized over all evaluable map cells. For region b, $L(b) = \sum a(g) \max[d(g),0]$ and $H(b) = \sum a(g) \max[-d(g),0]$. The common denominator is $N = \sum_b [L(b) - H(b)] = 0.1065143704$. The positive and negative percentages are $100\, L(b)/N$ and $-100\, H(b)/N$. They sum to +120.790563% and −20.790563%, respectively, and hence 100% net. Historical SST-association weights w(b,k) sum to one within each region; component amounts are $\sum_b L(b)w(b,k)$ and $-\sum_b H(b)w(b,k)$. Local subcomponents replace, rather than duplicate, their parent allocation. The global reference is not an estimate of each grid cell's expected AUC. The map aggregate differs from main Fig. 1's globally pooled monthly spatial-AUC deficit (0.1090497372); these are distinct estimands, with different ranking samples and aggregation order. Exact closure verifies accounting, not causal attribution. Source tables provide regional and component crosswalks to main Fig. 4.

Supplementary Table 1 | Data coverage

Figures	Variables, sources and periods	Spatial support, samples and leads
S1, S16–S17	SST; conditional intensity RMSE; pooled temporal AUC ERSST + retained four-model SST extraction Q90:1985–2014; maps:April–September next year; S17 also July–June	Ocean ≤60°; same-basin 11° neighbourhood; inset masks ±20° S1a:10 events; S1b:404 months; maps:3 events ×18 months ×9 leads Lead labels 1–9; single-class/invalid pools missing; original extraction limits retained
S2–S4	Monthly spatial AUC; Niño3.4 Same archived four-model SST ensemble as main Fig. 1 Monthly record; nine historical events + focal event	Global evaluation mask S2: phase-eligible months; S3: 9 events; S4: 1,080 records Leads 1–9; 12 verifying months per event; no independent-month inference
S5	SST, zonal wind stress, convection correlations NASA GEOS-S2S members 1–4; ERSST and ERA5 DJF 1997–98, 2015–16, 2023–24; climatology 1992–2020	20°S–20°N, 140°E–100°W; 4,833 common ocean cells 3 events × 3 variables × 8 leads = 72 correlations Exact starts and leads 1–8; lead 9 unavailable for all variables
S6–S9	200-hPa height; tropical precipitation for S9 ERA5; CCSM4, GEOS-S2S, SPEAR DJF; 1992–2020 monthly means, February 2018 excluded	S6/S8: all longitudes ≤60° incl. land; S7/S9: six ±23.5° recipient masks 3 events; 8 leads; S9: 144 repeated pairs Exact leads 1–8; NASA lacks 9; CFSv2 historical Z200 absent from this source set

Supplementary Table 1 | Data coverage (continued)

Figures	Variables, sources and periods	Spatial support, samples and leads
S10–S11	Height anomaly patterns, amplitude, MSE Same exact-coordinate three-model/ERA5 products DJF and common anomaly baseline	S10a global; S10b/c: 30–60°N, 40–60°S, 120°E east to20°E; S11:45°N,50°S 3 strong events; S10a additionally 6 other El Niño winters Means of exact leads 1–8; six reference winters:1994,2002,2004,2006,2009,2018
S12–S13	SST; indices; lagged SST/MLT; heating; advection ERSST, climate-index series and ORAS5 1985–2020; month effects and linear trend	Seven disjoint ±23.5° analysis regions 432 months/region; three forward test blocks No forecast lead axis; contemporaneous observed predictors explicitly diagnostic
S14	CFSv2 height and net surface heat-flux errors Separate CFSv2 process download; ERA5/ORAS5 verification DJF; normalization by two earlier event RMSEs	Seven ±23.5° regions; fixed common finite ocean cells 3 events × 7 regions × 3 leads; 54 remote pairs Leads 1,3,6; four members; not an independent model from main SST ensemble
S15	Local AUC differences and historical SST-association weights Same spatial AUC map and association weights as main Fig. 4 April 2023–September 2024; weights:1985–2020	Seven ±23.5° regions + Other seas; 11° same-basin pools 27,508 valid map centres; nine leads; weights:432 months/region Undefined AUC excluded; net map denominator 0.1065143704; distinct from Fig. 1 global AUC gap